# Ibrahim-Anyass Yussif

*Curriculum Vitae*

## PERSONAL DETAILS


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|---|---|
| *Citizenship* | Ghanaian |
| *Birth* | August 5, 1999 |
| *Address* | Cruise O'Brian Street, Dept. of Physics, University of Ghana |
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## PROFILE

Bachelor of Science in Physics with specialization in Biomedical Physics. Data analyst and air quality person with relevant experience in fieldwork. Experienced in using remote sensing tools and software such as ENVI, R, ERNDAS to perform statistical, spatial analysis and modeling. Data analyst and environmental consultant. Expertise in reference grade and low-cost air quality sensor assembly, deployment, maintenance, evaluation and calibration. Team player, result oriented, self-drive, highly motivated, smart and hungry to learn new technologies, methodologies, strategies and processes.

## EDUCATION

**Bachelor of Science in Physics with Biomedical Physics** — Sept 2018 - 2022
*College of Sciences, Department of Physics, Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana*

**West African Senior School Certificate Examination** — 2014 - 2017
*KNUST Senior High School, Boadi-Kumasi, Ghana*

## RESEARCH INTERSTS

*Atmosphere and climate Dynamic*

- Atmospheric Physics and air pollution
- Land-human-ocean-atmosphere and climate interactions
- Remote sensing and its applications in land and ocean's surface and climate modeling
- Regional climate-ocean dynamics and extreme events

*Environmental Health Science*

## WORK EXPERIENCE

**Teaching Assistant** — October 2022 - 2023
*University of Ghana (Department of Physics)*

- Supervised level 100 laboratory work
- Assisted laboratory technicians in setting up instruments and preparing for sessions
- Assisted faculty with classroom instruction and activities Mentoring student

**Field Assistant**

*University of Ghana ("Pathways to Equitable Healthy Cities" Pathways project)* October 2023 - 2024

- Use programming languages such as R, to detect outliers and perform time series analysis Identify and secure air and noise monitoring sites
- Assemble and deploy monitors per sampling protocols ensuring quality control measures
- Operate and maintain field monitoring equipment without interruption
- Apply good documentation practices when recording data and managing source documentation
- Implement quality control procedures in terms of downloading, storing and managing data

**Field Assistant**

*University of Ghana ("Indoor Air Quality monitoring in Selected Homes in Accra", project)* June to August 2024

- Checking air quality in homes using Zefan, and Upas monitoring instruments.
- Checking data for accuracy and integrity and resolving problems
- Use programming languages R, to detect outliers and perform time series analysis
- Writing good documentation practice when recording and managing source documentation.

**Data Analyst** January 2024 - Present

*University of Ghana (Afri-SET: "The Air Sensor Evaluation and Training Centre for West Africa", project)*

- Collect and analyze datasets of different types from low-cost air sensors, reference grade monitors, satellite products, reanalysis and model simulations using R and Python Software
- Mentor and provide technical guidance to students, interns, and junior researchers on air quality modelling and data analysis
- Assist in conducting public outreach activities to educate people about risks associated with certain activities and behaviors of air pollution
- Co-Author in the preparation of publications including policy briefs, reports as well as publish in high-quality peer reviewed national and international journals
- Data-driven discovery of climate dynamics using Machine Learning
- Participate in relevant conferences, workshops, training programs for a comprehensive understanding of developments relevant to air quality issues

## PUBLICATIONS

1. *James Nimo, Mathias A. Borketey, Emmanuel K.-E. Appoh, Abena Kyerewaa Morrison,* ***Yussif Ibrahim-Anyass****, Audrey Owusu Tawiah, Raphael E. Arku, Selina Amoah, Esi Nerquaye Tetteh, Tim Brown, Albert A. Presto, R. Subramanian, Daniel M. Westervelt, Michael R. Giordano, and Allison Felix Hughes (2025). "Low-cost PM2.5 Sensor Performance Characteristics against Meteorological influence in sub-Saharan Africa: Evidence from the Air Sensor Evaluation."* ***https://pubs.acs.org/doi/abs/10.1021/acs.est.4c09752***

2. *James Nimo,* ***Ibrahim-Anyass Yussif****, Mathias A. Borketey, Emmanuel K-E Appoh, Benjamin Essien, Selina Amoah, Joanna Modupeh Hodasi, and Allison F. Hughes. (2025) "Space-Time Air Quality Disparities in Sub-Saharan Africa: $PM_{2.5}$, $PM_{10}$, and Black Carbon Trends in the Greater Accra Metropolitan Area." https://doi.org/10.5194/egusphere-egu25-9303*

3. ***Yussif Ibrahim-Anyass,*** *James Nimo, Mathias A. Borkety, Selina Amoah, Allison F. Hughes. "Performance Evaluation and Application in Environmental Monitoring: Cross-sectional study utilizing Low-Cost Electrochemical Gas Sensors"* ***-In preparation***

## TECHNICAL SKILLS

***Software***

R, ERDAS, ENVI, Python, ArcGIS, Latex, Mesoscale Atmospheric Model (WRF, MM5),

**Weather and climate instrumentation & data analysis**

- Air Quality Sensors (AirGradient, Praxis, IQAir etc.), Reference Equivalent monitors (Teledyne etc)
- Davis weather instrument, polar-orbiting satellite (MODIS, MERRA-2)
- Remote sensing and its applications in land and ocean's surface and climate modeling

## ADDITIONAL EDUCATION, PRESENTATION & WORKSHOP

Presentation (ACCEPTED)
"***Assessing space-time disparities of fine particulate matter (PM2.5) levels in two urban settings of Ghana: Empirical evidence from calibrated low-cost air quality sensors***"
European Aerosol Conference 2024, ID: 430. **(Co-authored**)

***"Temperature, Precipitation, & More: Datasets for Comprehensive Analysis of Local Climate Change Impacts"***
**(USEPAZoom, July 2024)**

**"CAFE University - Wrangling & Processing Historic Weather Data for Climate and Health Studies" (Zoom, July 2024)**

**Certification in Air Quality Management and Science Training**
*(University of Ghana, ISSER Conference Center, 2024)* ***(Workshop, January 2024)***

Certificate of Completion in "**Carbon Dioxide Removal (CDR) and Solar Radiation Modification (SRM)"** short course
Department of Physics, University of Ghana, **(In Person, June 2024)**

Certificate of Participation in **"West African School on Air Quality and Pollution Prevention**" held from 4th to 16th November,2024 at Kwame Nkrumah University of Science and Technology (KNUST,2024 funded by Clean Air)

**ASAQ Online Seminar- "Air Quality data analytics and machine learning"** (November 26-28, 2024)

**"Future Changes in Extreme Precipitation Events Over the Volta Basin"** Dr. Jacob Agyekum, GMET, KNUST **(Virtual)**

**"Design and Fabrication of locally made Bolus material for radiotherapy purposes"** KNUST - Department of Physics **(POSTER)**

**"Remote Sensing Image Acquisition, Analysis and Applications" (UNSW Sydney, Coursera, 2023)**

Introduction to Data Analytics
***(IBM- Course, Coursera)***

**"Oceanography: a key to better understand our world" (Universität de Barcelona, Coursera, 2024)**

**Water Supply and Sanitation Policy in Developing Countries Part 2: Developing Effective Intervention**
*(University of Manchester, https://www.coursera.org/account/accomplishments/certificate/42KJYCX6G8PX )*

## REFERENCES

Available Upon Request.